

1: How much would \$5,000 in 35 years be worth today if the discount rate were 5.0%?

2: You want to buy a new ski boat 2 years from now, and you plan to save \$7,500 per year, beginning one year from today. You will deposit your savings in an account that pays 7.2% interest. How much will you have just after you make the 2nd deposit, 2 years from now?

3: You want to quit your job and return to school for an MBA degree 3 years from now, and you plan to save \$5,000 per year, beginning immediately. You will make 3 deposits in an account that pays 5.9% interest. Under these assumptions, how much will you have 3 years from today?

4: What's the present value of \$925 discounted back 8 years if the appropriate interest rate is 6%, compounded monthly?

5: Suppose Community Bank offers to lend you \$13,000 for one year at a nominal annual rate of 10.75%, but you must make interest payments at the end of each quarter and then pay off the \$13,000 principal amount at the end of the year. What is the amount of each payment?

6: Your father is about to retire, and he wants to buy an annuity that will provide him with \$69,000 of income a year for 25 years, with the first payment coming immediately. The going rate on such annuities is 5.70%. How much would it cost him to buy the annuity today?

7: Your grandmother just died and left you \$35,000 in a trust fund that pays 6.8% interest. You must spend the money on your college education, and you must withdraw the money in 4 equal installments, beginning immediately. How much could you withdraw today and at the beginning of each of the next 3 years and end up with zero in the account?

8: At a rate of 7.5%, what is the present value of the following cash flow stream? Years: 0 1 2 3 4
CFs: \$0 \$85 \$255 \$0 \$340

9: What's the present value of \$10,500 discounted back 5 years if the appropriate interest rate is 4.3%, compounded semiannually?

10: Suppose Community Bank offers to lend you \$13,000 for one year at a nominal annual rate of 10.75%, but you must make interest payments at the end of each quarter and then pay off the \$13,000 principal amount at the end of the year. What is the effective annual rate on the loan?

11: You plan to borrow \$36,100 at an 8.5% annual interest rate. The terms require you to amortize the loan with 7 equal end-of-year payments. How much interest would you be paying in Year 2?

12: How long will it take for a \$25,000 deposit in a bank account with an annual interest rate of 12% to double in value? Assume the interest is compounded annually.

13: How long will it take for a \$25,000 deposit in a bank account with an annual interest rate of 12% to double in value? Assume the interest is compounded monthly.

14: A perpetuity pays \$2,000 annually, with an interest rate of 7%.

- a. What is the present value of this perpetuity if payments are made at the end of each year?
- b. What is the present value of this perpetuity if payments are made at the beginning of each year?